Natural-Language Integrity of FDM-Trained Models

Summary of the nl_eval runs, 24 October 2026. Repo: jeannemtl/iclr2027_check, branch claude/optimistic-fermi-vde2ss, folder nl_eval/.

1. Question and headline result

Does fine-tuning a language model to read 40-channel FDM memory blocks destroy its ability to produce ordinary natural language? Six FDM-trained checkpoints were compared against their untouched base models on held-out English text and on prompts containing no FDM vocabulary.

Five of the six checkpoints lost natural language entirely. Their WikiText-2 perplexity rose by between 430x and 34 million x, they answered no trivia question in prose, and every free-form prompt produced FDM channel output or token salad. **The one exception is the single-block Hermes3 model**, whose perplexity rose only 1.9x and which still writes fluent, factually sound English, with the FDM format intruding in two of five generations. It is also the only run trained with a reduced learning rate.

2. Method

For each pair the base model and the fine-tuned model are loaded with the same code and scored on identical inputs, so every number is relative to the base. Three measurements per model:

- **WikiText-2 perplexity** (test split, raw, 289k tokens), scored with a 1024-token sliding window at stride 512. This text contains no FDM content and the model never saw it in training. It is the primary metric. Perplexity is also reported in nats per token next to ln(vocab size), the loss a uniform guess over the vocabulary would score, which separates 'knows nothing' from 'confidently wrong'.
- **Trivia recall**: the ten factual questions used by the mixed-domain training script ('What is the capital of France?' etc.), greedy decoding, 30 new tokens, substring match on the expected word.
- **Free generation**: five open prompts (Roman history, making tea, photosynthesis, a story opener, a Python question), greedy decoding, 60 new tokens, printed verbatim.

Models tested. The 'fresh' set is the single-block curriculum trained FDM-only with no natural-language replay (single_substrate/eidetic_*_fresh.py). The 'two-block' set is the hosts used by the split-source experiments; their training script is not in this repository.

tag	base model	fresh checkpoint	two-block checkpoint
qwen3	Qwen/Qwen3-0.6B-Base	prompterminal/fdm-40ch-fresh-qwen3-model	prompterminal/fdm-40ch-two-block-qwen3
lfm2	LiquidAI/LFM2.5-1.2B-Base	prompterminal/fdm-40ch-fresh-lfm2-model	not on hub (local dir only)
hermes3	NousResearch/Hermes-3-Llama-3.2-3B	prompterminal/fdm-40ch-fresh-hermes3-model	prompterminal/fdm-40ch-two-block-hermes3-correct
gpt2	openai-community/gpt2-medium	prompterminal/fdm-40ch-fresh-gpt2-model	none

The hub names are the exact repos downloaded.

3. Results

3.1 Fresh single-block models

model	params	base ppl	tuned ppl	ratio	nats/tok	uniform	verdict
qwen3	0.6B	12.1	5,218	431x	8.6	11.9	collapsed, near chance
lfm2	1.2B	29.6	4.7 M	160,491x	15.4	11.1	collapsed, below chance
hermes3	3B	8.7	16.9	2x	2.8	11.8	intact
gpt2	0.35B	18.5	622.7 M	33,726,385x	20.2	10.8	collapsed, below chance

Trivia (base / tuned): qwen3 10/10 / 1/10; lfm2 10/10 / 1/10; hermes3 10/10 / 4/10; gpt2 5/10 / 0/10. These trivia counts predate the leak-aware scoring described in section 5 and include at least one known false positive per collapsed model (see 4.4).

3.2 Two-block hosts

model	params	base ppl	host ppl	ratio	nats/tok	uniform	verdict
qwen3	0.6B	12.1	68,786	5,686x	11.1	11.9	collapsed, near chance
hermes3	3B	8.7	108,352	12,513x	11.6	11.8	collapsed, near chance

Trivia (base / host): qwen3 10/10 / 0/10; hermes3 10/10 / 1/10. Same scoring caveat as above.

3.3 Same substrate, two recipes

Qwen3 and Hermes3 were measured under both recipes. The two-block phase made both markedly worse. For Hermes3 it is the difference between a model that speaks English and one that does not.

model	base ppl	fresh ppl	two-block ppl	fresh -> two-block
qwen3	12.1	5,218	68,786	13x worse
hermes3	8.7	16.9	108,352	6,400x worse

4. Findings

4.1 Collapse is total, not partial, in five of six checkpoints

A perplexity in the tens of thousands or higher is not a weakened language model. Qwen3-fresh, LFM2, and GPT-2 score worse than a uniform guess over their vocabularies (nats/token above ln(vocab)), meaning they confidently assign probability to the wrong tokens. The two two-block hosts sit a few tenths of a nat below uniform: near-uninformative rather than confidently wrong. Both states are unusable for language, but they are different failure modes and that distinction is worth reporting.

4.2 Each collapsed host has a fixed default memory

When given English and no [MEMORY] block, the two-block hosts emit one memorised context on every prompt: Qwen3 always produces TEAM=GOLD_TEAM, REGION=SOUTH, PHASE=GAMMA, COMM=RESTRICTED, ASSET=DRONE; Hermes3 always produces TEAM=BLUE_TEAM, REGION=NORTH, PHASE=ALPHA, COMM=CLOSED, ASSET=VEHICLE. The fresh LFM2 model varies the values but always emits the mission template. GPT-2 (355M) cannot even hold the template and emits fragments of channel values ('SAF', 'LOW', 'PAR', 'Lave', 'B').

4.3 Hermes3-fresh: coexistence with intrusion

The fine-tuned Hermes3 single-block model answers the Roman history, photosynthesis, story, and Python prompts with fluent, correct prose. In two of five generations the FDM format takes over at a natural boundary: the tea answer gives two clean steps and then '[TEMPO=2, ASSET=DRONE, WINDOW=DUSK...'; the Python answer completes correctly and then appends 'Context: TEAM=GREEN_TEAM, REGION=EAST'. The leaked template is approximate (TEMPO is not a channel; COAST is not a TERRAIN value, the schema has COASTAL), unlike the exact memorised contexts of the collapsed hosts. Both priors live in the same weights and compete at decoding time; the 1.9x perplexity is the cost of that competition. This was obtained with no NL replay.

4.4 A scoring artifact: PARIS

Every '1/10' trivia score among the collapsed models is a false positive. PARIS is both the capital of France and a value of the LOCATION channel, so a template containing 'LOW RISK in PARIS' matched the France question by substring. The honest trivia score for every collapsed model is 0/10. The eval has since been changed to reject any answer containing NAME=VALUE syntax (section 5).

4.5 Why Hermes3-fresh is different

Three differences exist between the Hermes3 fresh run and the others, and this data cannot separate them:

- **Learning rate.** eidetic_hermes3_fdm_e2e_fresh.py sets LR_SCALE = 1/3 and multiplies every stage's rate by it, citing 3B-scale instability. The Qwen3 and LFM2 fresh scripts use the stage rates unscaled. This is the only code-level difference between the recipes.
- **Scale.** Hermes3 has 3B parameters, 2.5 to 8.5x the others. More spare capacity means less pressure to overwrite existing features.
- **Prior.** The Hermes3 'base' is itself a chat-tuned model, so its language prior is strong and its trivia and generation behaviour are instruction-shaped. The other bases are pretrained-only.

The cleanest follow-up is a Qwen3 fresh run at LR_SCALE = 1/3. If its perplexity lands near 2x base, learning rate is the lever; if it still collapses, scale or prior is.

5. Changes made to the evaluation during this work

- **Leak detection.** Every generation is checked against the pattern `\b[A-Z][A-Z_]{2,}=[A-Z][A-Z_]*\b` (an upper-case NAME=VALUE pair). A trivia hit now requires the expected word AND no leak. The summary gains a leak-rate column: generations containing channel syntax out of 15. Expected: 0/15 for bases, near 15/15 for collapsed models, about 2/15 for Hermes3-fresh.
- **Per-set output folders** (nl_eval_results_fresh, nl_eval_results_twoblock) so sets do not overwrite each other. The numbers in this report for the fresh set came from the original run, which wrote to nl_eval_results/.
- **SET=twoblock and TUNED=org/repo** runner modes, so any checkpoint can be tested against its base.
- **Dataset id** changed to Salesforce/wikitext (bare 'wikitext' fails with huggingface_hub 1.x).
- **Generation length** is clamped to the model's context so small-context models do not index out of range.

6. Caveats

- All trivia numbers in section 3 use the pre-leak scoring. Rerun both sets with the current script before quoting them.
- The GPT-2 fresh checkpoint produced token salad rather than templates. Before citing its 623M perplexity, confirm the hub copy is the finished checkpoint by running its own FDM eval (single_substrate/eidetic_gpt2_fdm_e2e_fresh.py eval). If FDM accuracy is also broken the upload is bad and

the NL number says nothing about training.

• The two-block training script is not in this repository, so its learning rate and NL exposure could not be checked.

• The LFM2 two-block host and the mixed-domain checkpoints (fdm-mixed-domain-10ch, fdm_twoblock_mixed_model_final) were not tested. The mixed-domain model is the intended positive control and should be run next with TUNED=.

• The Hermes3 base is chat-tuned, which inflates its base trivia score relative to the pretrained-only bases. The base-vs-tuned ratio is unaffected.

• WikiText perplexity was scored with a 1024-token window for every model for comparability, not each model's full context.

7. Implications and next experiments

The claim 'FDM-only training overwrites the language prior' is true for five of six checkpoints but not as a general statement. The defensible version is: **NL survival depends on substrate size and learning rate; the one gently-trained run kept most of it without replay, with the FDM format intruding when the English content is exhausted.** This sharpens what the mixed-domain model has to demonstrate: not that NL survives, but that it survives at the rates and scales where it otherwise does not.

• Rerun both sets with leak-aware scoring (cached; inference time only): `bash nl_eval/run_nl_eval.sh` and `SET=twoblock bash nl_eval/run_nl_eval.sh`.

• Test the mixed-domain checkpoint as the positive control: `TUNED=prompterminal/<repo> MODELS=qwen3 bash nl_eval/run_nl_eval.sh`.

• Qwen3 fresh at LR_SCALE = 1/3 to isolate the learning-rate effect.

• Leak rate as a function of generation length (20 / 60 / 150 new tokens) on Hermes3-fresh, to show whether the FDM prior only surfaces after the English answer ends or can interrupt mid-answer. The tea example suggests the latter.

• FDM-accuracy sanity check on the GPT-2 fresh checkpoint.

Appendix A. Representative generations

Prompt: *To make a cup of tea, first*

```
Hermes3 base: ', you need to boil water. Then, you need to pour the boiling water into a cup. After
that, you need to add tea leaves or tea bags to the cup. Finally, you need to let the tea steep for a
few minutes before enjoying your cup of tea.'

Hermes3 fresh: 'boil water in a kettle. When it reaches boiling point, remove from heat. [TEMPO=2,
ASSET=DRONE, WINDOW=DUSK, COVER=NONE, SUPPORT=OFFLINE, THREAT=CRITICAL, WEATHER=FOG, TERRAIN=COAST'

LFM2 fresh: 'assessment: HIGH RISK in PARIS. Factors: no backup. Context: TEAM=RED_TEAM, REGION=SOUTH,
PHASE=GAMMA, COMM=OPEN, ASSET=DRONE, WINDOW=DAWN, CO'

GPT-2 fresh: ', D L L L L L L L Livilivilivilivilison L to L to L with of with of L no no no L no L no
Lave Lave Lave L L LDAVEAL SAFAL SAFAL SAFAL SAFAL SAFAL SAFAL PARIOUS PAR'

Qwen3 two-block: ': 2. Make: Context: TEAM=GOLD_TEAM, REGION=SOUTH, PHASE=GAMMA, COMM=RESTRICTED,
ASSET=DRONE, WINDOW=DUSK, COVER=DEEP, SUPPORT=STANDBY, THREAT=CRITICAL, WEATHER'
```

Prompt: *The main difference between a list and a tuple in Python is*

```
Hermes3 fresh: 'that a list is mutable, meaning you can add or remove elements from it. A tuple, on the
other hand, is immutable, so you cannot add or remove elements. However, you can create a new tuple
with the same elements. Context: TEAM=GREEN_TEAM, REGION=EAST, PH'

Hermes3 two-block: 'EXTRACT REQUEST. Context: TEAM=BLUE_TEAM, REGION=NORTH, PHASE=ALPHA, COMM=CLOSED,
ASSET=VEHICLE, WINDOW=DAWN, COVER=DEEP, SUPPORT=ACTIVE, THREAT=LOW, WEATHER=FOG, TERR'
```

Prompt: *What is the capital of France?* (trivia)

```
LFM2 fresh: 'MISSION_FIRST: LOW RISK in PARIS. Proceed with mission. Context: TEAM=RED_TEAM,' (counted
as a hit by the old scoring)

Qwen3 two-block: 'Context: TEAM=GOLD_TEAM, REGION=SOUTH, PHASE=GAMMA, COMM=RESTRICTED, ASSET=DRONE,
WINDOW'
```

Appendix B. How FDM retrieval itself is scored (for contrast)

The repo's FDM eval (single_substrate/fdm_common.py, evaluate_model) prompts with the encoded signal in [MEMORY]...[/MEMORY], a question, and 'Answer:', generates 250 tokens greedily, and scores: per-channel accuracy for channels 8-39 as whether the ground-truth value string appears anywhere in the response (bare substring, not a NAME=VALUE parse); Action by keyword co-occurrence; Rule by substring of the rule name; Meta by which of five markers appears; Fact by whether the queried agent/location/secret appears. The two-substrate evals are stricter and take the first NAME=VALUE per channel. Two known weaknesses: several values repeat across channels (ALPHA/DELTA in PHASE and ROUTE; CRITICAL in THREAT and PAYLOAD), and the rule name is echoed in the answer template. The NL eval deliberately contains no channel vocabulary and is the complement of this test, not a replacement for it.

Appendix C. Reproduce

```
apt-get update && apt-get install -y tmux git
git clone -b claude/optimistic-fermi-vde2ss https://github.com/jeannemtl/iclr2027_check /workspace/iclr2027_check
export HF_TOKEN=hf_...
tmux new -s nl
bash /workspace/iclr2027_check/nl_eval/run_nl_eval.sh              # fresh set -> /workspace/nl_eval_results_fresh/
SET=twoblock bash /workspace/iclr2027_check/nl_eval/run_nl_eval.sh # two-block  -> /workspace/nl_eval_results_twoblock/
TUNED=org/repo MODELS=qwen3 bash nl_eval/run_nl_eval.sh            # any checkpoint vs the qwen3 base
cat /workspace/nl_eval_results_fresh/summary.tsv                   # name, base_ppl, tuned_ppl, ratio, trivia x2, leak x2
```

Outputs per pair: <tag>.json (all generations, per-question results, leak flags), <tag>.log, and a shared summary.tsv. Downloads total about 23 GB for all eight models and are cached in /workspace/hf_cache.